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Correlates of physical activity and sedentary behavior in children and adolescents during school recess: a systematic review and meta-analysis

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Abstract

Background Physical activity (PA) and sedentary behavior (SED) during school recess in children and adolescents are associated with numerous factors, such as age, gender, and adult engagement. Identifying these correlates can inform the design of recess programs and interventions in schools, while also providing evidence to support the establishment of future recess-related policies. Therefore, this study aimed to identify the correlates of PA and SED in children and adolescents during school recess.

Methods A systematic search was conducted for studies published in PubMed, EBSCO, and Web of Science from May 2011 to December 2024. We conducted three rounds of literature screening, including removing duplicates, screening titles and abstracts, and full-text screening. This study followed the PRISMA guidelines. The risk of bias was assessed using the Cochrane risk of bias tool. The effect of interventions on PA and SED was assessed by random effects meta-analysis. Identified factors were categorized using the social ecological model.

Results 72 studies were included in this systematic review. Multiple factors, including age, gender, Body Mass Index (BMI), family economic status, length and frequency of school recess, structured recess, provision of exercise equipment, and playground markings, have been identified as being positively associated with PA among children and adolescents during school recess. Gender factors are also associated with SED. The meta-analysis included 6 studies. Overall results indicated that interventions implemented during school recess had an impact on Light PA (SMD = 0.53; 95% CI = 0.14, 0.93; I^2=83%), Moderate PA (SMD = 0.38; 95% CI = 0.16, 0.60; I^2=25%), Vigorous PA (SMD = 0.26; 95% CI = 0.09, 0.44; I^2=0%), and Moderate-to-Vigorous PA (SMD = 0.49; 95% CI = 0.09, 0.88; I^2=86%) in children and adolescents. However, there was no statistically significant effect of the intervention on SED (SMD=-0.54; 95% CI=-1.15, -0.06; I^2=91%) ($P > 0.05$).

Conclusion This review identified several individual, social, physical environment, and organizational and policy level factors that correlate with PA among children and adolescents during school recess. However, it found a correlation only between gender at the individual level and SED. Therefore, future research should focus more on the relationship

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between social, physical environment, and organizational and policy level factors and SED among children and adolescents during school recess.

Keywords School recess, Physical activity, Sedentary behavior, Children and adolescents, Correlates

Background

Higher levels of physical activity (PA) have been positively linked to the physical, mental, and cognitive health of children and adolescents [1]. Additionally, PA levels during childhood have a lasting impact on health in adulthood [2]. The World Health Organization (WHO) recommends that children and adolescents should engage in at least 60 min of moderate-to-vigorous PA (MVPA) every day [3]. However, according to the analysis of data from 146 countries and territories in 2019 by Guthold et al., over 81% of children and adolescents worldwide have not met the WHO PA guidelines [4]. As a result, improving PA levels for children and adolescents presents a significant challenge. Since children and adolescents spend much of their time in school, school plays a crucial role in promoting their PA levels. School recess, as an important part of daily school life, provides opportunities for children and adolescents to engage in PA [5]. Research indicates that steps taken during recess account for 17–44% of the total steps taken during the school day, contributing significantly to daily PA [6, 7].

Sedentary behavior (SED) is a risk factor associated with the development of various health issues, including obesity and metabolic syndrome [8]. Moreover, SED not only negatively impacts the physical and mental health of children and adolescents but also contributes to a growing public health burden on society [9]. Given these adverse effects, addressing SED has become a key priority in public health efforts. The Global Action Plan on Physical Activity (2018–2030), published by the WHO for the first time includes the reduction of SED as one of the strategies for the prevention and control of global chronic diseases [10]. Despite this, in recent years, there has been a global increase in the prevalence of excessive SED among children and adolescents [9]. Previous studies have shown that schools play a crucial role in reducing SED among children and adolescents and that school recess is an important period to interrupt SED [11].

Numerous factors are associated with PA and SED among children and adolescents during school recess. While previous research has systematically identified correlates of PA among children and adolescents during school recess, the associations between individual factors, organizational and policy factors, and the length of school recess with PA remain unclear [5]. Moreover, a systematic review of both PA and SED in conjunction during school recess remains limited in the literature. To enhance PA levels and reduce SED, it is crucial to comprehensively identify the correlates of PA and SED among children and adolescents during school recess. This will inform the design of effective recess interventions to enhance PA and reduce SED among children and adolescents during school recess. Therefore, this study aimed to identify the correlates of PA and SED in children and adolescents during school recess using the social ecological model.

Methods

This systematic review and meta-analysis was registered with the Prospective Register of Systematic Reviews (PROSPERO) (CRD42024572168) and conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [12].

Literature sources and search strategy

This research conducted a comprehensive search of published papers between May 2011 and December 2024 in three electronic databases (PubMed, EBSCO, and Web of Science). The search strategy for the various databases included the following search terms in five main areas: population (child, children, youth, adolescent, teenager, juvenile, young people); school (elementary school, primary school, junior school, middle school, senior school, high school, secondary school); recess (recess, break time, school recess, playtime, school lunch recess, free play); physical activity (physical activity, exercise, energy expenditure, caloric expenditure, aerobic activity, aerobic exercise); sedentary behavior (sedentary behavior, sedentary time, sedentary lifestyle, sitting time, prolonged sitting time).

Inclusion criteria and selection process

Inclusion criteria: (1) involved children and/or adolescents aged 5–18 years; (2) PA and/or SED as the outcome variable; (3) PA and/or SED identified specifically during school recess; (4) exploring associations between PA and/or SED and other factors; (5) published in English in journals; (6) published from May 2011 to December 2024; Previous review has identified the correlates of physical activity in children and adolescents during school recess [5], which had a deadline of April 2011, so May 2011 was used as the start date of the literature search for this review.

All studies identified by the search were imported into EndNote, where duplicates were removed before screening. In the initial stages of the literature screening process, two independent reviewers (KZ, XYY) conducted an initial screening of the titles and abstracts of the identified articles. Following this, a third reviewer (JJJ) compared the screening results and reviewed the full text of the articles with disagreements. Subsequently, two independent reviewers (KZ, XYY) comprehensively reviewed the selected articles based on the initial screening of titles and abstracts. Only articles meeting the inclusion criteria were included in the study. Disagreements regarding inclusion were resolved through discussion with a third reviewer (JJJ).

Data extraction

Data were extracted from the included articles into standardized tables by two reviewers (KZ, XYY). Once the extraction was completed, a third reviewer (ZWX) randomly sampled the data to check for consistency. The inclusion form contained the following information: author, year, country, study type, sample characteristics (age, gender, sample size), the assessed school recess period, measures of PA and/or SED, and the research results.

Risk of bias

To guarantee the standardization and scientific rigorousness of the study, two reviewers (KZ, JJJ) employed the Cochrane risk of bias tool for randomized and non-randomized studies to assess the risk of bias in the included studies [13]. The tool consists of seven questions that assess the risk of bias, categorizing the studies as "low risk", "unclear", and "high risk". The seven questions are as follows: (1) random sequence allocation, (2) allocation concealment, (3) blinding of participants and personnel, (4) blinding of outcome assessment, (5) incomplete outcome data, (6) selective reporting, (7) other sources of bias. Two reviewers assessed all articles included in the study. Disagreements regarding assessment were resolved through discussion with a third reviewer (XYY).

Coding classification

The interventions and correlates identified in the studies were categorized into four levels based on the social ecological model: individual, social, physical environment, and organizational and policy. Each study included in the systematic review was coded according to the procedure outlined in previous reviews, where the correlation of an identified factor with recess PA and SED was determined by the number of findings supporting the direction of correlation [5]. For example, if the reported results of the studies demonstrated that a factor was positively correlated with school recess PA or SED, this was coded as "+". If the correlation was negative, it was coded as "-"; if no correlation was observed, it was coded as "0"; and if the correlation was uncertain, it was coded as "?" (Table 1).

The final calculations were made for each of the influencing factors. A factor was defined as having no correlation "0" if the result of the calculation was between 0% and 33% in correlation with PA or SED in children and adolescents. If the result was between 34% and 59%, it was defined as unclear "?". If the result was between 60% and 100%, a positive "+" or negative "-" correlation was determined based on the primary outcome. The result was coded as "++" if the intervention and factor were studied four or more times and supported as a positive correlation; "- -" if a negative correlation; "00" if no correlation; "??" if the correlation was uncertain.

Table 1 Rules for classifying factors based on correlation with school recess physical activity and sedentary behavior [5]

Studies supporting correlation (%)	Summary code	Explanation of code
0–33	0	No correlation
34–59	?	Uncertain correlation
60–100	+	Positive correlation
60–100	-	Negative correlation

Results

A total of 12,215 studies were retrieved from three electronic databases (PubMed, EBSCO, and Web of Science). After removing duplicates, 123 studies were retained for inclusion following a review of their titles and abstracts. 72 studies were retained for inclusion in this study after a full-text eligibility assessment. Ultimately, 6 studies met the criteria for inclusion in the meta-analysis (Fig. 1).

Literature characteristics

Detailed characteristics of the studies that met the inclusion criteria are shown in Appendix Table 1. The majority of studies were conducted in American (n = 27) [7, 14–39] and the remaining studies were conducted in Spain (n = 10) [40–49], Belgium (n = 4) [50–53], Australia (n = 4) [54–57], France (n = 3) [58–60], Finland (n = 3) [61–63], Brazil (n = 3) [11, 64, 65], England (n = 2) [66, 67], Denmark (n = 2) [68, 69], Netherland [70], Mexico [71], Greece [72], New Zealand [73], South Africa [74], Germany [75], Ecuador [52], Italy [76], Welsh [77], Slovenia [78], Qatar [79], Poland [80], Japan [81], China [82], and Canada [83]. 61 studies were conducted in primary schools and only 11 studies were conducted in secondary schools, with sample sizes ranging from 6 to 3919 individuals. Measurement of PA and SED in the included studies was mainly through the following: ActiGraph, Accelerometer, Pedometer, Multi-Sense Wear Armband, System for Observing Play and Leisure Activities in Youth (SOPLAY), Physical Activity Questionnaire for Children (PAQ-C), Physical Activity Questionnaire for Adolescent (PAQ-A), System for Observing Play and Recreation in

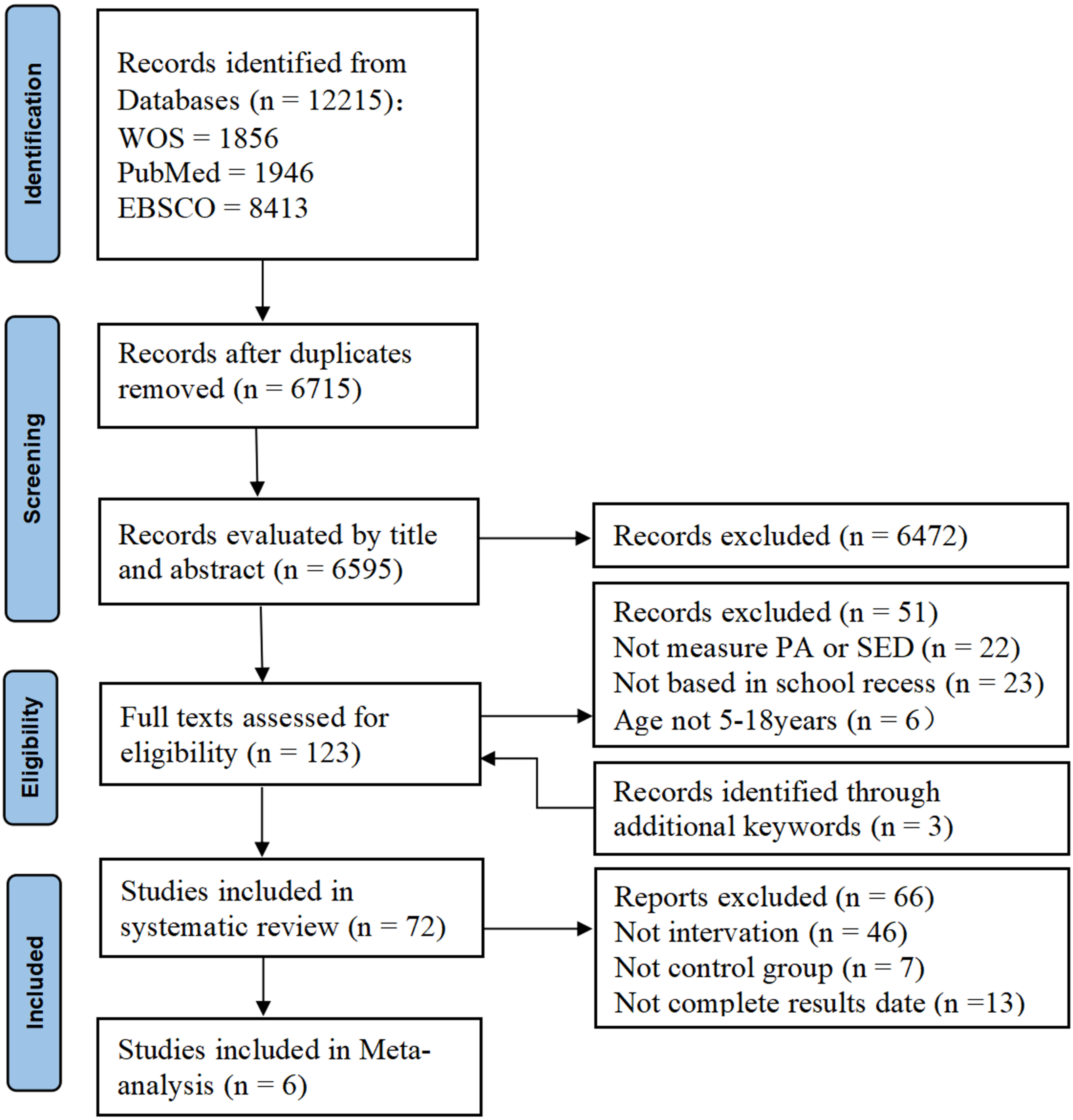

Fig. 1 Evidence inclusion and exclusion process

Communities (SOPARC), Self-reference questionnaire, System for Observing Children's Activity and Relationships during Play (SOCARP), Scale of physical activities, Subscale of activities at recess, Fitbit Flex, PANIC Physical Activity Questionnaire, Recess Physical Activity Recall (RPAR), Physical Activity During Recess Observation Questionnaire (PADROQ). The 72 studies included 9 randomized controlled trials, 4 non-randomized controlled trials, 41 cross-sectional studies, 15 quasi-experimental studies, 2 prospective cohort studies, and 1 comparative study.

Risk of bias assessment

The results of the risk of bias assessment for each study were summarized in Appendix Fig. 1 and Appendix Fig. 2.

Correlates of physical activity and sedentary behavior during school recess

Tables 2 and 3 summarize the correlations of PA and SED among children and adolescents during school recess. For consistency and comparability with previous reviews, correlates that were identified four or more times were reported below [5].

Correlates of physical activity

Individual factors The association between thirteen individual factors and PA was identified in 19 studies. Among them age [22, 40, 45, 48, 56, 60, 71, 77, 81], gender [14, 17, 22, 31, 40, 42, 45, 47, 48, 60, 71, 77, 79], and BMI [36, 40, 42, 47, 75, 79] were studied four or more times. The findings of the study indicated that there were significant differences in PA levels between ages. Furthermore, boys had higher PA levels than girls during school recess. Previous reviews have not identified an association between BMI and PA, whereas the present study found a correlation between them.

Social factors In 13 studies, six social factors associated with PA were identified, but only family economic status [40, 48, 58, 71] was studied four times. The study results indicated a correlation between family economic status and PA levels during school recess.

Physical environment factors In 28 studies, twenty-two physical environmental factors associated with PA were identified, including playground markings [18, 35, 59, 60] and providing equipment [16, 18, 23, 29, 35, 37, 43, 52, 56, 62, 72, 82], which were identified four or more times. The findings indicated that the implementation of playground markings and the provision of entertainment equipment during school recess increased PA levels in children and adolescents.

Organizational and policy factors In 17 studies investigating the association between fourteen organizational and policy factors and PA, length and frequency of school recess [25, 31, 33, 75] and structured recess [30, 49, 50, 62, 76] were identified four or more times. Structured recess is associated with recess PA in children and adolescents. The present study found a positive correlation between children's and adolescents' PA levels and school recess length and frequency.

Correlates of sedentary behavior

Individual factors In 9 studies investigating the association between nine individual factors and SED, gender, age, and BMI were identified multiple times, but only gender was identified more than four times [11, 47, 58, 60, 63, 64, 73]. The study's results indicated that boys had less SED than girls.

Relevant studies investigating the association between social factors and SED, which meet the inclusion criteria have not yet been collected. Furthermore, there are fewer studies on the association between organizational and policy factors and physical environmental factors and SED. Despite numerous factors being investigated, each factor has only been identified once.

Meta-analysis

A total of 6 studies [23, 30, 53, 54, 59, 76] were included in the meta-analysis, and 8 findings from these studies were included: Light PA, Moderate PA, Vigorous PA, Very High PA, Very Vigorous PA, Light Vigorous PA, MVPA, and SED. Seven interventions were identified for their effects on children's and adolescents' PA levels and SED: playground marking, structured recess, staff training (specialized training for teachers involved in school recess activities), provision of sports equipment, staff training combined with the provision of exercise equipment, sport education-based irregular teaching units, and traditional sports education. Meta-analysis of the effects of the interventions on Very High PA, Light Vigorous PA, and Very Vigorous PA was not performed, as only one of the studies that met the inclusion criteria reported effects on these outcomes.

The results of the Meta-analysis showed that the intervention affected Light PA (SMD = 0.53; 95% CI = 0.14, 0.93; I^2=83%) (Appendix Fig. 3), Moderate PA (SMD = 0.38; 95% CI = 0.16, 0.60; I^2=25%) (Appendix Fig. 4), Vigorous PA (SMD = 0.26; 95% CI = 0.09, 0.44; I^2=0%) (Appendix Fig. 5), and MVPA (SMD = 0.49; 95% CI = 0.09, 0.88; I^2=86%) (Appendix Fig. 6). The effect of the intervention on SED was not statistically significant (SMD=−0.54; 95% CI=−1.15,−0.06; I^2=91%) (Appendix Fig. 7).

Sensitivity analysis

Interventions implemented during the school recess increased Light PA, Moderate PA, Vigorous PA, and MVPA in children and adolescents. We performed sensitivity analyses by sequentially excluding studies, which revealed no significant change in the I^2 for Moderate PA and Vigorous PA, but significant changes in Light PA and MVPA. This suggests a considerable amount of variability between the studies included in the meta-analysis for Light PA and MVPA.

Publication bias

Egger's test was employed to assess publication bias, and the results indicated no significant bias, with P-values as follows: Light PA (P = 0.643), Moderate PA (P = 0.119),

Table 2 Factors correlated with physical activity during school recess

Variables	Found correlation		No correlation	Summary code	
	Reference no.	**Correlation (±)**	**Reference no.**	**n/N (%)**	**Correlation**
Individual factors					
Age	[22, 40, 45[a], 56, 60, 71, 77, 79, 81]	-	[46]	10/11 (91)	--
	[45[b]]	+			
Gender	[14, 17, 22, 31[c], 40, 42, 45, 47, 48, 58, 60, 71, 71, 79]	+	[16, 34, 46]	14/17 (82)	++
Race			[16, 17]	0/2 (0)	0
Grade	[7, 17]	-		2/3 (67)	-
	[48[a]]	+			
BMI/Weight	[36[d], 40, 42, 47, 48, 75]	-	[7, 36[e]]	6/8 (75)	--
Social margination index	[71]	-		1/1 (100)	-
Goal setting	[20, 22, 26]	+		3/3 (100)	+
Self-monitoring	[26]	+		1/1 (100)	+
Peer relationships school	[61]	+		1/1 (100)	+
Pro-social behavior	[32]	+		1/1 (100)	+
Parental educational level	[63]	+	[47]	1/2 (50)	?
Sports Participation	[65]	+		1/1 (100)	+
Social factors					
Staff training	[16, 18, 23]	+		3/3 (100)	+
Public posting			[20]	0/1 (0)	0
Rewards/feedback	[26]	+	[20, 22]	1/3 (33)	0
Adult engagement	[31, 32]	+		2/2 (100)	+
Teach/adult supervision	[31, 74]	+		2/2 (100)	+
Family economic status	[40, 58, 63, 71]	+		4/4 (100)	+
Physical environmental factors					
Playground Markings	[12, 18, 35, 59]	+	[54]	4/5 (80)	++
Playground size	[72, 82]	+		2/2 (100)	+
Playground crowding	[74]	-		1/1 (100)	-
Playground	[73]	+		1/1 (100)	+
Playground density	[51]	-		1/1 (100)	-
Self-constructed material	[41]	+		1/1 (100)	+
School infrastructure	[14, 71, 83]	+		3/3 (100)	+
Green elements/schoolyard renovation	[19, 21]	+		2/2 (100)	+
Music	[43]	+		1/1 (100)	+
Physical education-based intervention			[35]	0/1 (0)	0
Provide equipment	[16, 18, 23, 29, 35, 37, 38, 43, 52, 56, 62, 72, 78, 82]	+		14/14 (100)	++
School climate	[61]	+		1/1 (100)	+
Rainfall	[66]	-		1/1 (100)	-
Environmental Characteristics of School	[37, 55]	+		2/2 (100)	+
Walking Track	[67]	+		1/1 (100)	+
Schoolyard Renewal	[68]	+		1/1 (100)	+
Recess Dance Videos	[34]	+		1/1 (100)	+
Play area	[46]	+		1/1 (100)	+
Seasonal changes	[57[e]]	-		1/1 (100)	-
Temperature	[21]	-		1/1 (100)	-
Shade	[21]	+		1/1 (100)	+
Space available	[80]	+		1/1 (100)	+
Organizational/policy factors					
Structured recess	[30, 44, 49[h], 50[f], 62]	+	[5]	5/6 (83)	++
Unstructured recess	[15]	+		1/1 (100)	+

Table 2 (continued)

Variables	Found correlation		No correlation	Summary code	
	Reference no.	Correlation (±)	Reference no.	n/N (%)	Correlation
length and frequency of school recess	[25, 31, 33, 75]	+		4/4 (100)	++
Ban on smartphone usage during recess	[69]	+		1/1 (100)	+
Good Behavior Game (GBG)	[24]	+		1/1 (100)	+
Application of the specific program	[76]	+		1/1 (100)	+
Playwork	[27[g], 39[b]]	+	[39[a]]	2/3 (67)	+
Recess Enhancement Program (REP)	[28]	+		1/1 (100)	+
School break time policies	[30]	+		1/1 (100)	+
Playgrounds program	[70]	+		1/1 (100)	+
Sport Education based irregular teaching unit	[53]	+		1/1 (100)	+
Traditional Sport Education			[53]	0/1 (0)	0
Reducing restrictions on bringing in sports equipment to school	[56]	+		1/1 (100)	+

Table 3 Factors correlated with sedentary behavior during school recess

Variables	Found correlation		No correlation	Summary code	
	Reference no.	Correlation (±)	Reference no.	n/N (%)	Correlation
Individual factors					
Age	[8, 11[b]]	-		2/3 (67)	-
	[45]	+			
Gender (male)	[8, 11, 15, 33, 36, 45, 64]	-		7/7 (100)	--
Family economic level	[36[a]]	-	[13, 45]	1/3 (33)	0
BMI/Weight status	[8]	+	[35[e], 45]	1/3 (33)	0
Screen time			[45]	0/1 (0)	0
Ethnicity			[15]	0/1 (0)	0
Inadequate eating habits	[8]	+		1/1 (100)	+
Waist circumference	[8[b]]	+		1/1 (100)	+
Parental educational level	[36[a]]	-		1/1 (100)	-
Physical environmental factors					
Self-constructed material	[36]	-		1/1 (100)	-
Playwork	[12[b]]	-		1/1 (100)	-
School type	[8[b]]	+		1/1 (100)	+
Playground Markings	[26]	-		1/1 (100)	-
Playground density	[28]	-		1/1 (100)	-
Rainfall	[61]	+		1/1 (100)	+
Organizational/policy factors					
Structured recess	[58]	+		1/1 (100)	+
Recess frequency	[31]	-		1/1 (100)	-

The summary code provides an overall summary of the findings for each variable. "0", no correlation; "00", no correlation for an outcome that has been studied four or more times; "?", uncertain correlation; "??", uncertain correlation for an outcome that has been studied four or more times; "+", positive correlation; "++", positive correlation for an outcome that has been studied four or more times; "-", negative correlation; "--", negative correlation for an outcome that has been studied four or more times. a: Boy; b: Girl; c: There were more males than females at the age of ten; d: School morning recess; e: School lunch recess; f: MVPA; g: Vigorous PA; h: Children's lunch recess physical activity levels decreased between winter, spring and summer

Vigorous PA ($P=0.531$), MVPA ($P=0.247$), and SED ($P=0.308$). All P-values were greater than 0.05, indicating that there was no publication bias in this study.

Discussion

This review aimed to identify the correlates of PA and SED in children and adolescents during school recess. Based on the socio-ecological model, the review categorized the correlates and interventions identified in the study. The findings indicated differences in PA levels during school recess based on children's and adolescents' age, gender, Body Mass Index (BMI), and family economic status. Gender is associated with SED in children and adolescents during school recess. Although more studies have been conducted since the previous review, few have identified the association between the social, physical environment, and organizational and policy factors and SED during school recess. Additionally, recess interventions have been shown to effectively increase PA levels among children and adolescents. 6 studies were included in the meta-analysis, and the overall results indicated that implementing interventions had an effect on Light PA, Moderate PA, Vigorous PA, and MVPA among children and adolescents during school recess.

Correlates of individual factors

At the level of individual factors, age, gender, and BMI are significant factors influencing PA and SED among children and adolescents during school recess. Studies have shown that PA levels during school recess decline with age [77]. As children and adolescents age, increased academic pressure leads to a decrease in the frequency and length of participation in PA, resulting in lower levels of PA and higher SED [84].

Regarding gender, boys generally have higher levels of PA than girls, while the opposite is true for SED. During school recess, boys were more inclined to participate in strenuous sports and team-based competitions, thus accumulating more PA [58]. In contrast, girls spend more time socializing and engaging in more SED [58]. Additionally, differences in activity areas may also contribute to differences in PA levels between boys and girls. Boys prefer activities such as soccer and basketball on the playground, while girls are inclined to favor swings and balance areas for their activities [85].

BMI is an important factor influencing PA levels during school recess among children and adolescents. Those with a normal weight tend to engage in more MVPA compared to their obese peers. Peer children and adolescents tend to perceive obese children and adolescents as having poorer motor skills and being destructive and aggressive, which makes it difficult for them to participate in group activities [86]. Obese children and adolescents are often reluctant to participate in activities after being ostracized. Furthermore, during school recess, they engage in less vigorous exercise and take fewer steps compared to their normal-weight peers [42].

Correlates of social factors

At the level of social factors, many factors remain uninvestigated. Regarding the correlation between PA and SED during school recess and social factors among children and adolescents, only family economic status was studied four times. Therefore, future research should focus on investigating the association between social factors and children's and adolescents' PA and SED to provide valuable information for schools to set up recess interventions.

There was an association between family economic status and children's and adolescents' PA levels during school recess. Children and adolescents from families with higher economic status tend to engage in more extracurricular sports training, acquire more sports skills, and form good sports habits, thereby participating in more sports activities during school recess [1, 38]. Furthermore, parents in better-off families are more aware of the benefits of their children's participation in sports, actively encouraging their children to participate in PA during school recess as a result [87].

Correlates of physical environment factors

The physical environment is an important factor influencing PA and SED in children and adolescents during school recess, with increased playground markings and the provision of sports equipment being the most common recess interventions. The results of a systematic review showed that increasing playground markings effectively enhanced PA levels in children and adolescents, which is consistent with the results of a previous review [5]. Moreover, meta-analysis results indicated that increasing playground markings did not affect children's and adolescents' Light PA. The results of the meta-analysis are consistent with the findings of Baquet et al., who conducted a twelve-month follow-up study [60]. The study revealed that initial changes in Light PA at the beginning of the intervention were probably due to a "novelty effect". However, after twelve months, the "novelty effect" diminished, and Light PA levels remained unchanged.

This systematic review indicates that providing sports equipment significantly increases PA levels among children and adolescents [35, 56]. Children and adolescents in schools with adequate sports equipment are more active during school recess than those in schools with inadequate sports equipment [72]. Providing adequate sports equipment in schools can be an effective solution to the issue of children and adolescents being unable to engage in PA due to a lack of equipment [72].

Correlates of organizational and policy factors

The organizational and policy factors influencing PA and SED among children and adolescents during school recess include structured recess and the length and frequency of school recess. Studies have shown that structured recess is an effective strategy for increasing PA levels in children and adolescents. During structured recess, children and adolescents spend significantly more time engaging in PA and less time in SED [50]. Structured recesses attract more children and adolescents to participate in PA, especially those who are usually inactive during school recess [50]. Moreover, children and adolescents accumulate more MVPA in structured recess [30, 76]. Previous reviews had not found an association between the length and frequency of recess and PA levels of children and adolescents during school recess [5]. Nevertheless, this study indicated that both the length and frequency of recess had a significant impact on children's and adolescents' PA levels during school recess. Longer recess or increased recesses provide children and adolescents with more opportunities to engage in PA, thus accumulating substantially more MVPA [75]. Additionally, children and adolescents in schools with more frequent recesses had less SED and spent more time engaging in PA [33]. Currently, many countries do not have mandatory recess periods. Therefore, these findings can provide valuable reference for developing policies on the length and frequency of recesses.

Study strengths and limitations

To our knowledge, this review is the most recent and comprehensive study to identify the correlates of PA and SED among children and adolescents during school recess. Using the social ecological model, this study systematically categorized and discussed factors at various levels, providing valuable insights for the design of effective recess interventions in schools. Through a systematic review of 72 relevant studies, we identified numerous factors associated with PA and SED during school recess, as well as the effects of interventions in increasing PA levels and reducing SED. Despite the thorough analysis provided in this study, several limitations remain. Firstly, for consistency and comparability with previous reviews, correlated factors identified four or more times will be reported. While this approach enhances data reliability, it may have excluded some less explored correlates, potentially overlooking factors that could influence PA and SED during school recess. Secondly, most of the included studies were cross-sectional in design, and no longitudinal studies were identified, which prevented clarification of the causal relationship between the correlates and PA and SED. Thirdly, the majority of the included studies focused on elementary school children, with fewer studies involving secondary school students. This limits the generalizability of the findings, particularly for adolescents, and highlights the evidence gap in this area. Lastly, Meta-analysis results should be interpreted with caution because of the limited number of studies included in the meta-analysis, especially the small number of randomized controlled trials. In addition, these studies differed significantly in terms of intervention type, methodology, and study objectives, resulting in a high degree of heterogeneity in some of the findings.

Conclusions

Age, gender, BMI, and family economic status are correlates of PA among children and adolescents during school recess. Furthermore, the implementation of playground markings, provision of exercise equipment, structured recess, extended recess periods, and increased recess frequency has been shown to enhance PA levels among children and adolescents during school recess. This review identified only a correlation between gender at the individual level and SED. Therefore, future research should focus more on the relationship between social, physical environment, and organizational and policy level factors and SED among children and adolescents during school recess.

Abbreviations

BMI	Body mass index
WHO	World health organization
PA	Physical activity
SED	Sedentary behavior
MVPA	Moderate-to-vigorous physical activity

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-23948-x.

Supplementary Material 1

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Authors' contributions

ALL authors were involved in the selection of the topic. KZ drafted the manuscript and conducted the article screening and data collection with JJJ, XYY, and ZWX. YL, GYQ, and YCL reviewed and edited the manuscript. All authors have read and approved the final version of the manuscript and agree with the order of authorship.

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Data availability

The data used and/or analyzed in the current study are available from the first author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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